Syed Muhammad Awais Raza

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AI/ML Engineer with production-ready FastAPI and Docker experience from O2Geeks AI internship, working **on CV, NLP,** and **RAG systems**. Built a full-stack AI marketplace with Next.js and pgvector-based recommendations, and developed YOLOv8-based cricket ball/bat tracking solutions. Founder of a **250+** member AI/ML society, delivering hands-on training from Python fundamentals to end-to-end MLOps pipelines

EXPERIENCE

AI/ML Intern | O2Geeks AI Program | Jul 25–Aug 29, 2025 | ML Engineering Lead

- Completed an intensive deep learning internship focused on **CV and NLP** (OCR, AR, semantic search) using **pgvector-based RAG systems**.
- Built and deployed **production ML pipelines** using **FastAPI, Streamlit, and Docker** under the guidance of the CEO and AI Research Lead.
- Deployed 3 production ML services using FastAPI and Docker.

EDUCATION

COMSATS University Islamabad, Attock Campus

B.S. in Artificial Intelligence *Expected Graduation, June 2027*

- **Current CGPA:** 3.4/4.00
- **Related Coursework:** Python Developer, Linear Algebra, Statistics, ML Fundamentals, Deep Learning, Computer Vision, NLP, MLOps Foundations

Technical SKILLS

- **Languages:** Python, Java (OOP/DSA), JavaScript (Next.js), SQL
- **Backend & Data:** FastAPI, Flask, PostgreSQL (SQLAlchemy), Redis
- **ML / DL:** PyTorch, TensorFlow, scikit-learn; MLflow, DVC
- **NLP:** spaCy, NLTK/Gensim (TF-IDF, LDA), VADER
- **MLOps / DevOps:** Docker, CI/CD (GitHub Actions/Jenkins), Airflow; Kubernetes (working knowledge)
- **Visualization:** Streamlit, Plotly, Matplotlib
- **Cloud & Tools:** Railway, TensorFlow Serving; GitHub, Jupyter Notebook, VS Code

Key PROJECTS

Campus Exchange (Founder/Lead Developer) | GitHub

- Built **AI-powered campus marketplace** using Next.js, FastAPI, PostgreSQL (pgvector), Redis with **15% pricing** accuracy gain and personalized recommendations.
- Deployed a **full-stack production system** addressing student scams and expense sharing across verified users.

Cricket Ball & Bat Detection + Speed Measurement (YOLOv8n) | Github

- Trained **YOLOv8 models** for ball–bat detection and hit events using bounding-box overlap and **Ultralytics tracking**.
- Implemented **3D trajectory reconstruction** via **Tomasi–Kanade** for real-time speed estimation.

Voice-ID & Clone: Integrated Speaker Identification and Neural Voice Cloning System |Github

- Developed a robust speaker identification engine using SpeechBrain ECAPA embeddings and a custom-calibrated cosine similarity threshold **(0.4814)** to accurately authenticate enrolled users against unknown queries.
- Designed **multi-stage audio pipeline** with **Coqui TTS** and automated preprocessing (OGG/MP3/WAV → 16kHz).

Heart disease, Bank churn prediction using ML models and different neural networks | Github

- Used Boosting algorithms and pipelines for better accuracy also used MLP's and FNN for prediction

ACTIVITIES AND LEADERSHIP

Hexagon AI/ML Society(Founder, 200+ members) *July 2024 –Present*

- Teach Python→ML→MLOps roadmap via blogposts (Markdown+code) and live sessions.
- Mentor international clients on production ML pipelines and FastAPI deployment.

Microsoft Learn Student Ambassadors (MLSA) (Tech Team Member, 2024-25)

- Recognized with Certificate of Appreciation for exceptional Tech team contributions during MLSA 2024-25 tenure.
- Supported campus-wide AI/ML workshops and technical events under Microsoft Learn program.

CERTIFICATIONS & LICENSES

- Python for Data Science – Anaconda (2024)
- Career Essentials in Generative AI – Microsoft & LinkedIn (2024)